

1. Execution Summary

This report presents a comprehensive analysis of the delivery operations of a global e-commerce company managing end-to-end order fulfilment across multiple regions. The analysis covers 172,765 orders spanning January 2015 through 2018, focusing on identifying root causes of chronic late deliveries, quantifying their financial impact and establishing a data driven framework for improvement.

The headline finding is stark: 54.71% of all orders are delivered late, costing the business \$2.1M in eroded profitability on delayed orders alone. While total profit across fulfilled orders reached \$7.5M, the persistent late delivery problem represents both significant financial drain and a major customer experience risk. A Random Forest predictive model achieved 74% accuracy in identifying at-risk orders before they become late.

Key Finding	Metric
Overall late delivery rate	54.71%
On-time delivery rate	45.29%
Total orders analysed	172,765
Total profit (profitable order)	\$7.5M
Profit at risk (delayed orders)	\$2.1M
Predictive model accuracy	74%

2. Business Context & Objectives

The company operates a global e-commerce platform selling products across categories including sporting goods, fitness equipment, outdoor gear, footwear and apparel across multiple international regions.

Core Business Problem

Actual shipping times frequently deviate from scheduled delivery windows, creating eroded customer trust,

Reduced order profitability and an inability to make reliable commitments to buyers at point of purchase.

Analytical Objectives

- Understand the current state of delivery performance across all dimensions (regions, mode, time, segment).

- Quantify the financial impact of delays on order profitability.
- Identify the primary operational bottlenecks driving late deliveries.
- Build a predictive model to flag high-risk orders before they are shipped.
- Deliver actionable recommendations to improve on-time delivery and profitability.

3. Key Performance Indicators (KPI)

The following KPIs were computed directly from the cleaned dataset to establish a performance baseline.

172,765 Total Orders	94,523 Late Deliveries	54.71% Late Delivery%	45.29% On-Time Delivery%
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\$7.5M Total Profit	\$2.1M Profit at Risk	3 Days 90th Percentile Delay	\$22.03 Average Order Profit
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Late Delivery Rate at 54.71%	More than half of all orders arrive late. This is not an edge-case problem it is the default experience for majority of customers.
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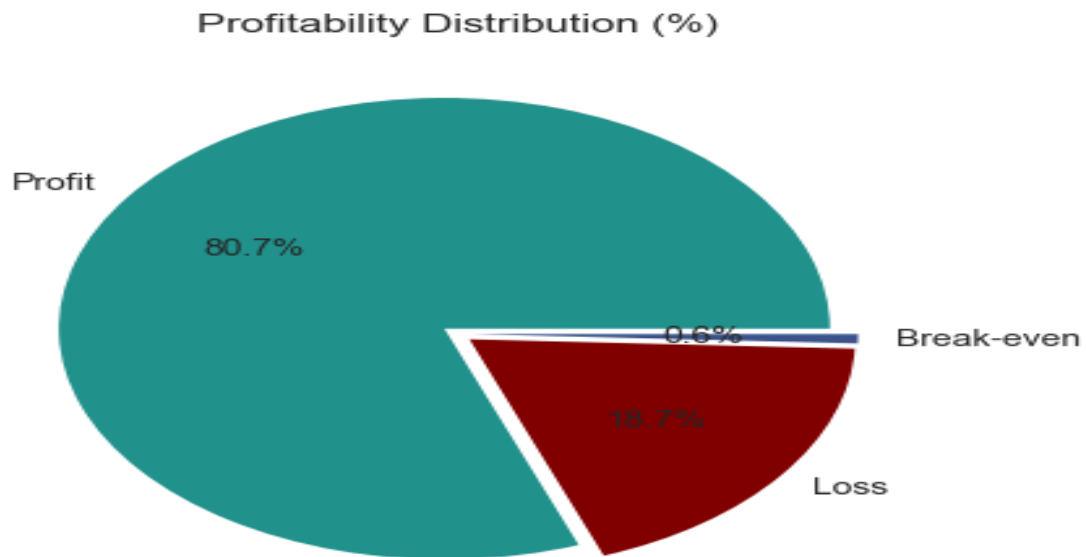
\$2.1M Profit at Risk	Orders that experienced delays collectively generated \$2.1M in profit that is under constant pressure from further operational deterioration.
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90th Percentile Delay = 3 Days	Even the most extreme cases are contained within 3 days of lateness, suggesting the problem is systemic and process-driven rather than catastrophic.
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4. Profitability Analysis

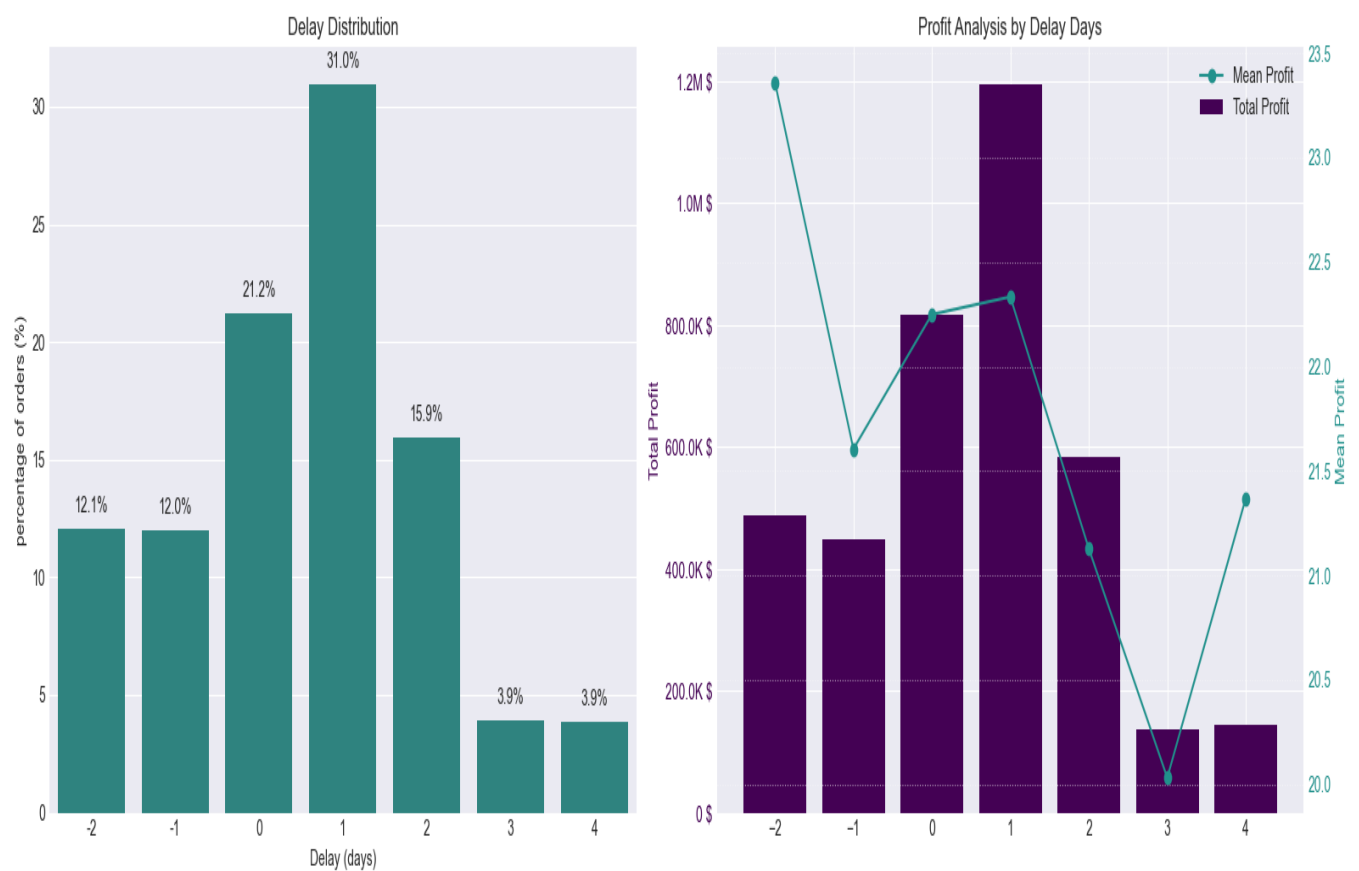
4.1 Profitability Distribution

Order-level profitability was classified into three tiers based on Order Profit Per Order. While 80.7% of orders are profitable, the 18.7% loss-making share represents a meaningful drag that is disproportionately concentrated among delayed shipments.



4.2 Delay Distribution & Profit vs Delay Days

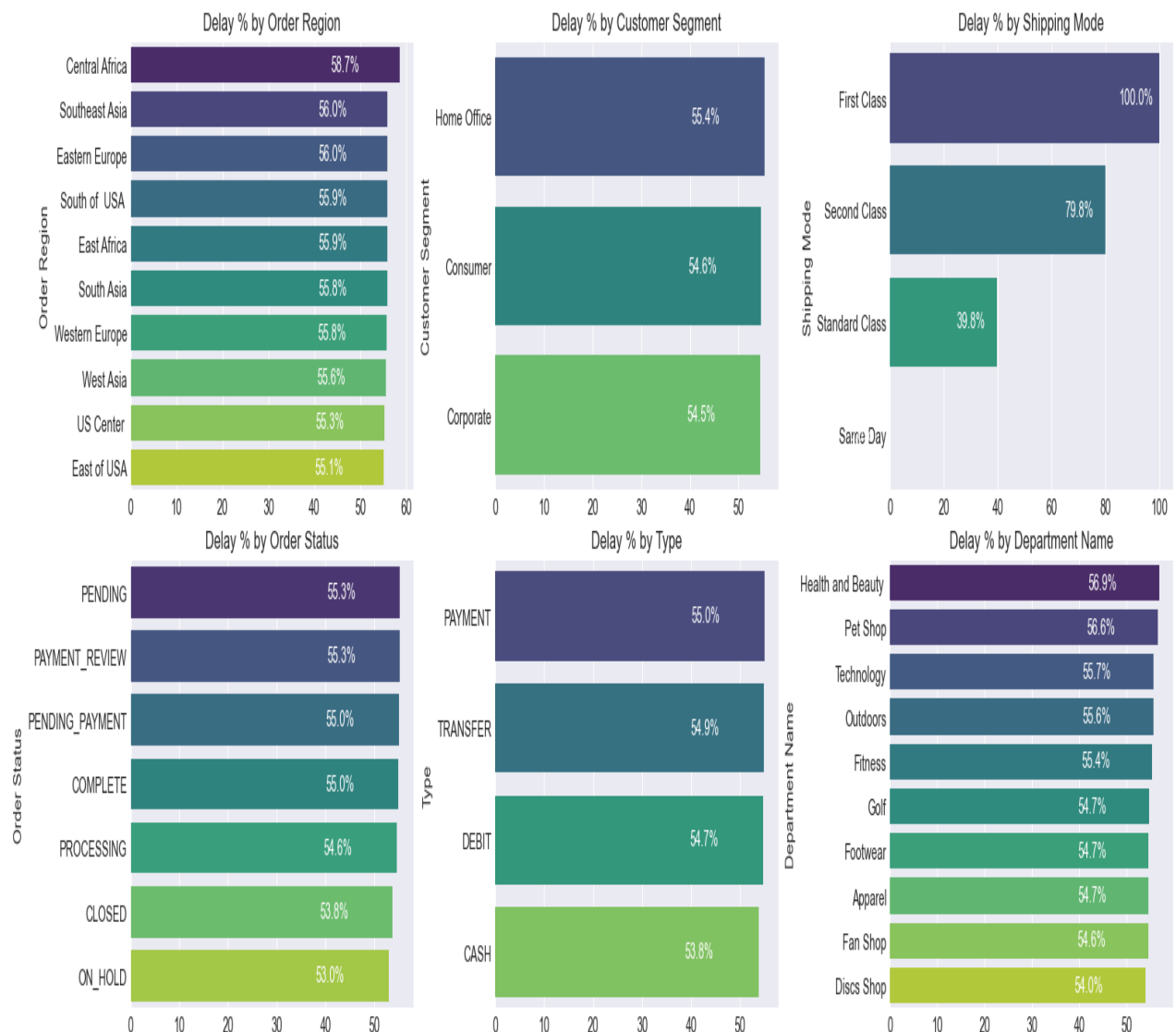
The delay distribution shows that 31.0% of all orders arrive exactly 1 day late the single largest cohort. Combined orders delayed by 1-4 days account for 54.7% of all order volume, directly mapping to the overall late delivery rate.



A critical observation: mean profit per order remains stable at approximately \$21-\$23 across all delay levels. The profitability problem is therefore driven by volume of delayed orders, not by individual order economics making systematic throughput improvement the highest-leverage intervention.

5. Bottleneck Detection

Delay percentage was computed across six key operational dimensions. The charts below reveal where the fulfilment process breaks down most severely.



Shipping mode is the # 1 Lever

First Class: 100% delay rate, Second Class: 79.8%. Standard Class: 39.8%. Same Day: 0%. The mode assignment is most logical.

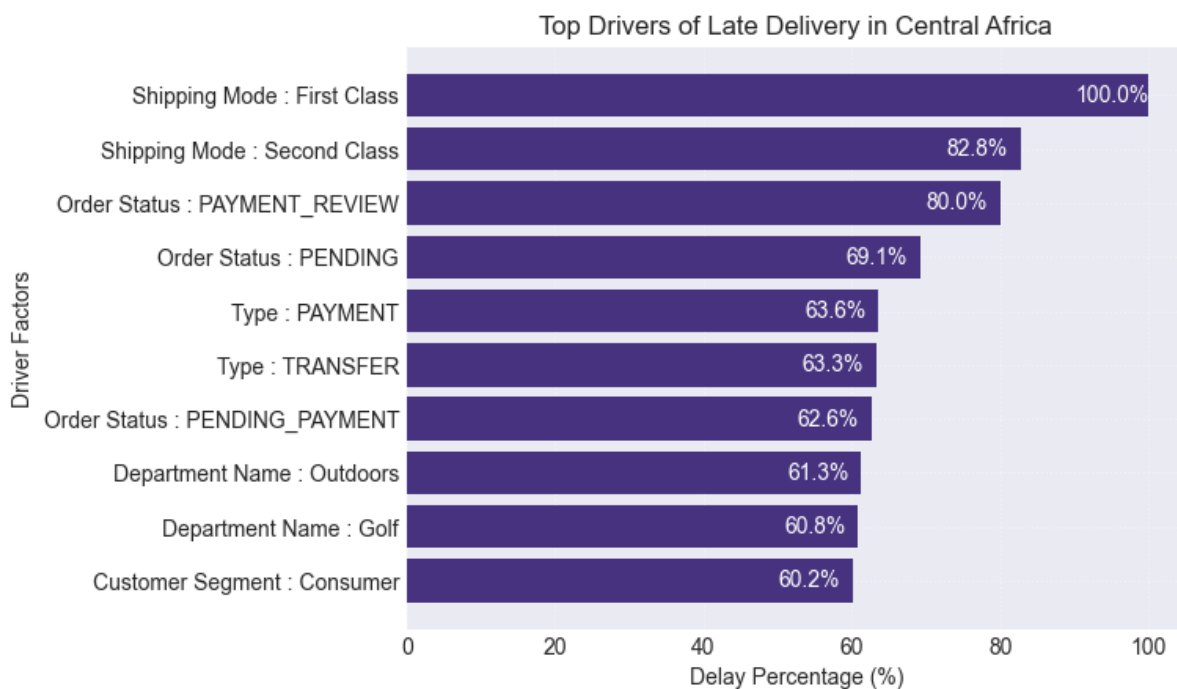
Regional Spread is Narrow (55-59%)	Central Africa leads at 58.7% but all regions cluster between 55-59%. This rules out a localised logistics failure and confirms a company-wide issue.
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Customer Segments Identical (54.5-55.4%)	No segment receives preferential service. All customers – Consumer, Corporate, Home Office – exp. The same broken delivery.
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Health & Beauty (56.9%) & Pet Shop (56.6%) Lead by department	These departments marginally outpace others in delay rate, warranting an inventory and carrier audit for these product categories.
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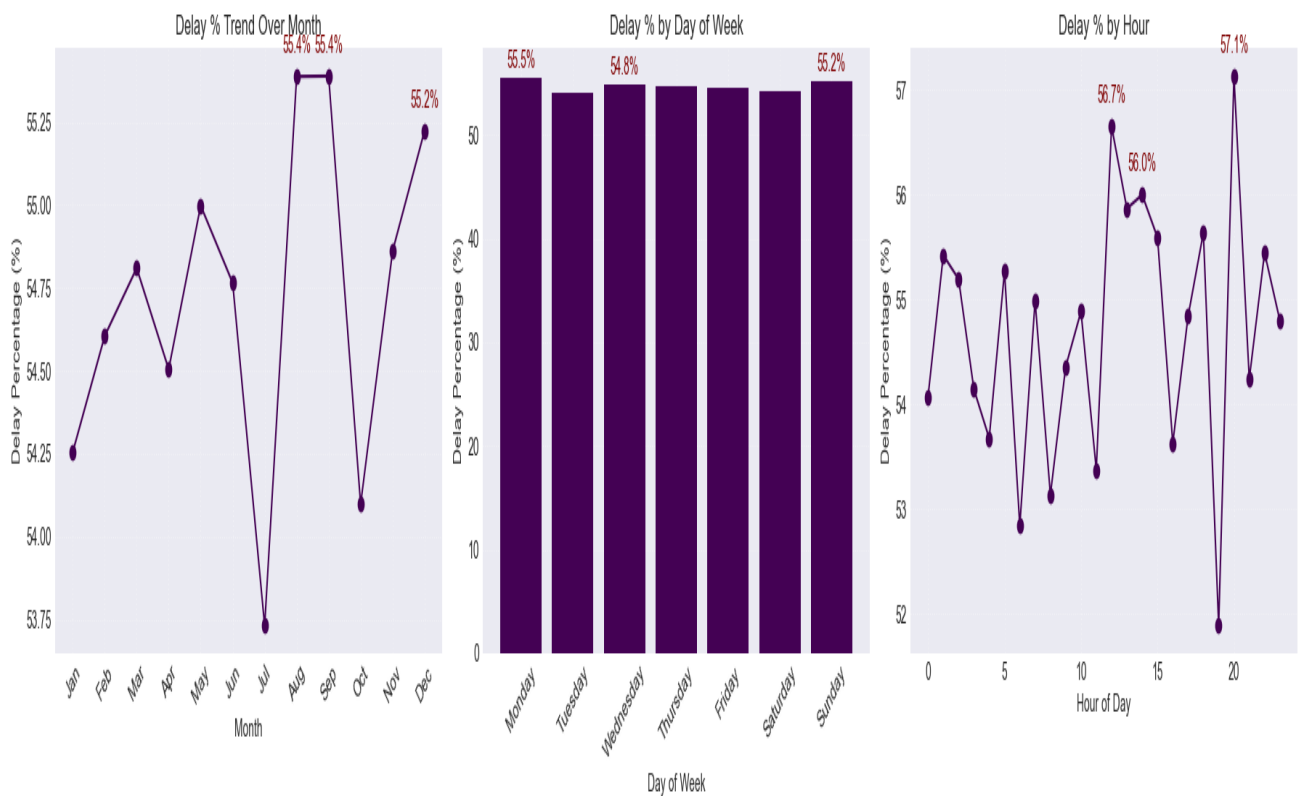
6. Root Cause Analysis

Central Africa was selected for deep-dive root cause analysis as the highest-delay region (58.7%). The chart below ranks the top 10 driver factors by delay percentage within this region.



7. Time-Based Delay Patterns

Three temporal dimensions were analysed: month of year, day of week and hour of day. While delay rates are relatively stable across all dimensions (53-57%), specific peaks highlight opportunities for targeted capacity planning.



Peak Delay Months: August & September (55.4%) and December (55.2%)	Aug & Sep are the highest months at 55.4%, with Dec close behind. These reflect mid-year promotions and Q4 holiday surge overwhelming fulfilment capacity.
Day of week Variance is Minimal	Monday (55.5%) and Sunday (55.2%) are the peak days, but the spread across the week is less than 1 percentage point. Day of week is not a good level.

8. Machine Learning Model Results

Class Imbalance Handling (SMOTE)

Split	Class 0 (On - Time)	Class 1 (Late)
Before SMOTE (train)	59,030	79,182
After SMOTE (train)	79,182	79,182

Random Forest Classifier – Test Set Performance (34,553 records)

Metric	Class 0 (On-Time)	Class 1 (Late)	Overall (Weighted)
Precision	0.68	0.78	0.74
Recall	0.72	0.75	0.74
F1-Score	0.70	0.77	0.74
Accuracy			0.74
Support	14,758	19,795	34,553

- The model correctly classifies nearly 3 out of 4 orders - a strong foundation for production deployment as an order level late delivery system.
- When the model predicts the order will be late, it is correct 78% of the time
- The model captures 75% of actual late orders.

